How To Interpret Photutils Optimiser Output

Reading the summary and detail CSV files from optimise_photutils_global_parameters.py

Main idea. The optimiser is not trying to maximise the amount of masked light. It is trying to find the least aggressive global Photutils settings that still cover bar-profile spike contamination while doing little damage to clean control galaxies.

What The Command Tests

The command evaluates every combination of the supplied nsigma, dilation, max-area, and max-elongation values. With 5 nsigma values, 3 dilation values, 3 max-area values, and 3 max-elongation values, it tests 135 parameter sets. For each parameter set it evaluates the default spike-positive galaxies and default no-spike control galaxies, then writes a ranked summary and a per-galaxy detail table.

Input option	Meaning	Interpretation
--nsigmas	Detection threshold in residual-sigma units.	Lower values detect fainter objects but risk false positives.
--dilations	Radius used to grow retained masks.	Larger values catch source wings but damage more nearby galaxy light.
--max-areas	Maximum retained segment area in pixels.	Larger values allow broader detections but risk masking galaxy structure.
--max-elongations	Maximum retained segment elongation.	Larger values keep more stretched detections; smaller values reject arms and streaks more strongly.

Where The Output Goes

Because the example command does not specify --pc or --output-dir, the script uses DEFAULT_PC = "Laptop". On this machine that means the default output folder is:

Default output path. C:\Users\gordo\Dropbox\Public Documents\UCLAN\MSc Research\Remove foreground objects\optimisation

If you want the run to write to the D:\Dropbox tree instead, use --pc Desktop or pass an explicit --output-dir.

The Two CSV Files

File	Purpose	How to use it
photutils_parameter_optimisation_summary.csv	One row per parameter combination, sorted with the best score first.	Start here. Use it to identify the top few candidate parameter sets.
photutils_parameter_optimisation_details.csv	One row per galaxy per parameter combination.	Use this to diagnose why a top-ranked or suspicious parameter set did well or badly.

Summary CSV Columns

Column	Meaning	Good direction
nsigma, dilation, max_area, max_elongation	The tested Photutils parameter values.	No universal direction; judge through the score and diagnostics.
smooth_sigma	Gaussian smoothing scale used to build the galaxy model before residual detection.	Fixed at 15 px in this command unless --smooth-sigma-pixels is supplied.
npixels	Minimum connected pixels required for a detection.	Fixed at 8 in this command unless --npixels is supplied.
central_exclusion	Central radius protected from source masking.	Fixed at 12 px in this command unless --exclude-center-radius-pixels is supplied.
mean_spike_coverage	Mean fraction of detected spike-profile samples covered by the mask in spike-positive galaxies.	Higher is better; values near 1.0 mean the spikes are covered.

Column	Meaning	Good direction
mean_control_damage	Mean profile-damage score in no-spike controls.	Lower is better; this is one of the key safeguards against over-masking.
mean_control_mask_fraction	Mean fraction of image pixels masked in control galaxies.	Lower is better.
mean_spike_mask_fraction	Mean fraction of image pixels masked in spike-positive galaxies.	Lower is usually better, provided spike coverage remains high.
score_lower_is_better	Weighted combined score used to rank parameter sets.	Lowest is best, but always inspect the top few visually.

How The Score Is Built

The score is a weighted penalty. It strongly penalises missing spike samples, then adds penalties for control profile damage and unnecessary mask size:

Score formula. score = (1 - mean_spike_coverage) x 10 + mean_control_damage x 5 + mean_control_mask_fraction x 2 + mean_spike_mask_fraction

This means missed spike coverage matters most. However, once several parameter sets cover the spikes well, the best choice should usually be the more conservative one: less control damage, less masked area, higher nsigma, smaller dilation, and smaller max_area where possible.

Detail CSV Columns

Column	Meaning	Use
group	Whether the row is a spike galaxy or a control galaxy.	Separate success cases from damage checks.
name	Galaxy name.	Identify which galaxy drives a good or bad score.

Column	Meaning	Use
spike_samples	Number of spike-profile samples detected in that galaxy.	Controls may have zero or few spikes; spike galaxies should have useful positive evidence.
spike_coverage	Fraction of spike samples covered for that galaxy.	Low values identify parameter sets that missed contamination.
segments	Number of retained Photutils segments.	Very high values can indicate aggressive false-positive detection.
masked_pixels	Number of pixels masked.	Useful for comparing absolute mask size across a specific galaxy.
masked_fraction	Fraction of the image masked.	Watch for unexpectedly high values, especially in controls.
profile_affected_fraction	Fraction of bar-major profile samples touched by the mask.	High values can mean the science profile is being heavily altered.
profile_damage	Combined log-profile change plus affected-profile penalty.	Use this to find over-masking in control galaxies.

A Practical Reading Workflow

Step	Action	Why
1	Open the summary CSV and sort by score_lower_is_better if it is not already sorted.	The script writes the best-ranked parameter sets first.
2	Look for mean_spike_coverage close to 1.0.	A low score is not scientifically useful if real spikes are missed.
3	Among high-coverage rows, compare mean_control_damage and mean_control_mask_fraction.	These tell you whether the settings harm clean galaxies.
4	Prefer the most conservative near-tie.	If scores are similar, choose less aggressive masking rather than the absolute lowest score.

Step	Action	Why
5	Use the detail CSV to inspect each top candidate galaxy by galaxy.	One bad control galaxy can be hidden by a good mean.
6	Generate visual reports for the top 3-5 candidates before adopting one.	Metrics narrow the field; visual inspection catches galaxy-structure mistakes.

Warning Signs

Pattern in output	Likely meaning	Response
High spike coverage but high control damage	The setting removes spikes but also damages normal profiles.	Try higher nsigma, smaller dilation, smaller max_area, or stricter max_elongation.
Low spike coverage and low mask fraction	The setting is too conservative to catch the contaminants.	Try lower nsigma, larger max_area, or slightly larger dilation.
Many segments in controls	The detection threshold or filtering is too permissive.	Raise nsigma or tighten max_area/max_elongation.
Best score uses very aggressive settings	The score may be dominated by spike coverage.	Compare near-ties and inspect detail rows plus visual reports.

Recommended Decision Rule

Choose this. A parameter set with near-complete spike coverage, low control damage, low control mask fraction, and conservative parameter values among the top-ranked rows.

Do not automatically choose the row with the absolute lowest score if a slightly higher-scoring row is visibly safer. For science-profile work, the best global Photutils setting is the one that removes foreground-object contamination without teaching the mask to remove galaxy structure.